

Foundry Work

6.2 PATTERN

Pattern is the model of casting. It is made of wood, metal or plastics. Mould is produced in moulding sand by using pattern.

6.3 PATTERN MATERIALS

The following materials are used for making patterns.

- Wood
- Metal
- Plaster
- Plastic
- Wax

Wood: Wood is widely used for making pattern. It is easy to work and readily available. Generally pattern is made from teak wood, mahogany, Pine and rose wood. The surface finish and life can be increased by applying metal coating on the wood pattern.

Metal: Metal pattern is used for producing large number of casting. Metal pattern is made by using a master pattern made of wood. Cast iron, Brass, Aluminium and white metal are commonly used for pattern making. Aluminium is the best metal for pattern making. Brass is suitable for small size patterns. White metal can be used for making patterns for complicated shapes.

Plaster: The gypsum cement is known as plaster. Pattern is made by pouring the mixture of plaster and water into the mould prepared by master pattern. It is used for making small patterns and core boxes.

Plastics: Plastic patterns are producing from a master pattern made of wood. The two types of plastics materials are used for pattern making namely, thermo setting and thermo plastics. Pattern made of thermo plastics used for producing less number of pattern. But Thermo setting plastics are used for large number of pattern.

Wax: Wax patterns are excellent for investment casting process. This pattern is produced from paraffin wax, shellac wax, bees wax, ceresinwax and micro crystalline wax. The liquefied wax is injected into a split die. Then the die is cooled and the wax pattern is taken out.

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